

M-TRACE

MPLADS Tracking, Risk, Anomaly, Compliance & Execution

AI-Powered Monitoring, Anomaly Detection & Decision-Support Platform for MoSPI

Problem ID: 26102

Department: DIID, MoSPI

Category: Software / Smart Automation

Core Administrative Principle: The biggest logical vulnerability is treating M-TRACE as if it will replace the existing MPLADS administrative process. It should not. M-TRACE is an intelligence, monitoring, evidence, and decision-support layer built around the existing process. Approvals, sanctions, tenders, payments, inspections, and corrective actions remain with the competent authorities.

1. Executive Summary

M-TRACE is a proposed AI-powered monitoring and decision-support platform designed to improve the monitoring of works implemented under the Members of Parliament Local Area Development Scheme (MPLADS).

MPLADS involves a large number of development works across constituencies. These works generate a substantial amount of information, including recommendations, sanctions, estimated costs, expenditure, payments, project progress, invoices, documents, photographs and completion information.

The fundamental problem is not simply the absence of data. The bigger problem is that large volumes of project and financial information are difficult to continuously analyze manually.

M-TRACE addresses this problem by bringing relevant project information into a unified monitoring environment and applying Artificial Intelligence, Machine Learning, Data Analytics, Anomaly Detection, Risk Scoring, GIS, Geo-tagged evidence, Satellite-assisted monitoring, Predictive Analytics, Compliance checking, Explainable AI, Automated alerts, Human verification, and Audit trails.

The platform does not automatically accuse anyone of fraud. Instead, when the system detects an unusual pattern, it generates an anomaly or risk alert, explains the reason behind it, identifies the supporting evidence, assigns a risk level, and recommends what should be reviewed. The final decision remains with the authorized government authority.

2. Official Problem Statement

ATTRIBUTE	DETAILS
Problem ID	26102
Title	Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation
Organization	Ministry of Statistics and Programme Implementation (MoSPI)
Department	Data Informatics & Innovation Division (DIID)
Category & Theme	Software Smart Automation

3. Background & Context

The Members of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme under which Members of Parliament recommend developmental works for creating durable community assets and providing basic civic amenities. A large number of works are undertaken across the country through different implementing agencies and administrative authorities.

A typical work involves an end-to-end lifecycle:

MP Recommendation → District-level Scrutiny & Processing → Sanction → Implementing Agency → Tender/Contract → Execution → Expenditure/Payment → Completion

Each project generates considerable information including project descriptions, locations, estimated costs, sanctioned amounts, implementing agencies, contractor details, tender records, expenditure figures, bills, invoices, progress milestones, photographs, documents, and completion certificates. When thousands of projects are implemented simultaneously, manually examining every data point becomes impossible. Therefore, the government requires an intelligent system to identify irregularities early.

4. The Actual Problem in Simple Words

Imagine thousands of MPLADS projects running across India. An authority has access to project, financial, and execution data, but simply having data does not answer critical operational questions:

- Is this project spending money unusually quickly?
- Is the project taking much longer than expected?
- Is the expenditure consistent with the physical progress?

- Is the estimated cost unusually high compared with comparable projects?
- Does the same or similar work appear to have been registered more than once?
- Is submitted evidence coming from the correct location?
- Are invoice values unusually different from relevant benchmarks?
- Is there a recurring pattern involving certain projects, locations or entities?
- Which projects should the authority examine first?

A human monitoring team cannot efficiently identify all such patterns manually at national scale. The core problem is how AI and advanced analytics can continuously examine MPLADS data to identify unusual patterns, risks, inefficiencies, and compliance deviations early enough for authorities to investigate.

5. Critical Reasons Why the Problem Exists

- **Huge Volume of Data:** Thousands of works generate massive financial and project information that manual examination cannot scale to handle.
- **Data Fragmentation:** Information resides across project records, financial logs, bills, images, GIS feeds, and progress reports without a unified integration layer.
- **Manual Monitoring Limitations:** Authorities cannot continuously inspect every project with uniform detail, necessitating a risk-based prioritization approach.
- **Hidden Patterns:** An individual record may look normal, but when compared against thousands of similar projects, deviations become apparent (e.g., a ₹50 lakh water tank cost that deviates significantly from comparable regional benchmarks).
- **Delayed Detection:** Anomalies discovered post-completion have limited utility compared to early warnings during execution.
- **Physical & Digital Gap:** Reconciling reported physical progress (**Progress = 70%**) with actual ground evidence requires connecting project data, financial records, GIS, and satellite evidence.

6. Proposed Solution & Core Philosophy

M-TRACE is a centralized AI-powered monitoring platform observing the complete lifecycle of MPLADS works to answer five major questions: *What is happening? (Tracking)*, *Is anything unusual happening? (Anomaly Detection)*, *How serious is it? (Risk Assessment)*, *Why was it flagged? (Explainable AI)*, and *What should authorities examine? (Decision Support)*.

Core Operating Cycle:

Monitor → Validate → Analyze → Detect → Predict → Alert → Verify → Act → Learn

7. Proposed User Roles & Jurisdiction

M-TRACE enforces strict role-based access control (RBAC) to ensure security and data privacy across administrative tiers.

ROLE	MAIN RESPONSIBILITY IN M-TRACE
MoSPI	National-level monitoring, macro analytics, risk trends, and high-level decision support.
State Authority	Monitoring MPLADS works and regional risk indicators within the state.
District Authority	Operational monitoring, physical verification, and project-level supervision.
MP / MP Office	Monitoring recommended works within their specific constituency.
Implementing Agency	Project execution information, milestone tracking, and implementation updates.
Contractor	Authorized submission of progress evidence, bills, and permitted project documents.

Important Note: Access is strictly jurisdiction-based. MPs view only their constituency works, contractors see only assigned projects, district authorities operate within their districts, and MoSPI maintains national visibility.

8. End-to-End System Workflow (Stages 1 to 17)

The system tracks projects across 17 structured stages, culminating in a unified data repository:

- 1. MP Recommendation:** Recording work name, purpose, location, GPS coordinates, estimated cost, expected beneficiaries, and timeline.
- 2. District Scrutiny:** District authorities verify required information, location validity, and documentation.
- 3. Sanction / Approval:** Recording sanctioned amount, sanction date, project ID, category, implementing agency, and expected completion date.
- 4. Implementing Agency Assignment:** Designating the responsible agency for project execution.
- 5. Contractor Assignment:** Associating the winning contractor with restricted project access to submit progress, photos, and bills.
- 6. Unique Project ID Generation:** Assigning a standardized identifier (e.g., MPLADS-MH-AMR-2026-000127) linking all lifecycle data.
- 7. Data Collection:** Aggregating project, financial, execution, geospatial, and photographic evidence.
- 8. Data Validation:** Automated checks for missing fields, invalid values, duplicates, format errors, date inconsistencies, and GPS errors. Flagging incomplete data as *Correction Required*.

9. **Unified Data Platform:** Centralizing validated project, financial, execution, document, image, GIS, and historical data into a comprehensive intelligence context.

9. AI & Analytics Engine Modules

The brain of M-TRACE consists of specialized intelligence engines operating on validated data:

- **Financial Intelligence:** Analyzes costs, expenditures, payments, invoice values, and material price benchmarks.
- **Execution Intelligence:** Monitors progress milestones, delays, and completion timelines.
- **Geospatial Intelligence:** Analyzes project coordinates, evidence coordinates, proximity, and GIS patterns.
- **Evidence Intelligence:** Screens photographs and documents for reuse, duplicates, inconsistencies, and potential AI/synthetic generation.
- **Compliance Intelligence:** Validates project records against predefined statutory rules and documentation requirements.

Key Anomaly Detection Mechanisms

- **Cost Anomaly:** Compares estimated costs (C_{est}) against historical and comparable benchmarks based on capacity, specifications, and location. Generates *Cost Anomaly — Review Required* upon significant deviation.
- **Expenditure vs. Progress Mismatch:** Detects anomalies where expenditure is disproportionately high relative to reported physical progress (Exp_{high} vs $Prog_{low}$).
- **Delay Detection:** Identifies timelines exceeding expected schedules and estimates delay probabilities using historical models.
- **Duplicate Work Detection:** Scans for overlapping work descriptions, identical locations, and duplicate project characteristics.
- **GPS Verification:** Compares approved project coordinates against geo-tagged progress photo coordinates, flagging *Location Mismatch* if tolerance thresholds are exceeded.

10. Satellite-Assisted Monitoring & Risk Scoring

Where suitable, M-TRACE integrates available satellite imagery to provide supporting spatial evidence of physical changes. Projects are visualized on a national GIS map using clear risk indicators:

● **Green — Normal** ● **Orange — Attention Required** ● **Red — High Risk**

Unified Risk Score & Explainable AI

M-TRACE aggregates multiple signals into a single standardized **Project Risk Score (0–100)**. Every alert incorporates **Explainable AI** answering four critical questions:

Explainable AI Framework:

- **WHY?** Why did the system flag the project?
- **WHAT?** What specific data points caused the alert?
- **HOW SERIOUS?** What is the calculated risk level?
- **WHAT NEXT?** What specific review actions should the authority take?

11. Case Management, Audit Trail & Governance

Serious anomalies transition into structured cases managed through the platform. Every action is recorded in an immutable **Audit Trail** (submission, modification, review, evidence upload, anomaly generation, investigation, finding, and final action).

Human Verification Loop: AI detects anomalies → calculates risk → explains reasons → alerts authority → authority reviews → requests additional evidence or field inspection → records finding. Outcomes are fed back into the system to refine models and reduce false positives over time.

What M-TRACE Does NOT Do:

M-TRACE does *not* automatically sanction projects, release funds, select contractors, cancel contracts, declare fraud, replace district authorities, replace physical inspections, replace financial procedures, or treat AI predictions as final decisions. It strictly provides evidence-based intelligence to support human authorities.

12. Complete M-TRACE System Architecture



13. Comprehensive Service Matrix

SERVICE MODULE	CORE PURPOSE & OPERATIONAL SCOPE
MPLADS Project Tracking	Track works across the complete lifecycle from recommendation to completion.
Financial Monitoring	Monitor estimated, sanctioned, and actual expenditure against allocations.
Payment & Invoice Analytics	Identify unusual payment patterns, billing discrepancies, and invoice values.
Cost Benchmarking	Compare project costs against comparable historical and regional benchmarks.
Progress & Delay Detection	Track physical progress milestones and identify delayed projects proactively.
Duplicate Detection	Identify potentially duplicated or overlapping project submissions.
GPS & GIS Monitoring	Compare project and evidence locations; visualize works geographically.
Satellite Monitoring	Provide supporting spatial evidence for physical asset changes where suitable.
Evidence Verification	Analyze photographs and documents for consistency and authenticity.
Anomaly & Predictive Analytics	Identify unusual patterns and predict future risks (cost escalation, delays).
Risk Scoring & Explainable AI	Prioritize projects based on risk scores (0–100) and explain why alerts occurred.
Smart Alerts & Dashboards	Route targeted alerts to MoSPI, State, District, MP, and Agency stakeholders.
Case Management & Audit Trail	Maintain end-to-end investigation trails, accountability, and compliance history.

14. Practical Illustration: Water Tank Project Lifecycle

Consider a realistic scenario demonstrating M-TRACE in action:

- **Recommendation:** MP recommends a 20-lakh-litre water tank in Village X with an estimated cost of ₹50 lakh.
- **Scrutiny & Sanction:** District authorities scrutinize and sanction the work through standard administrative procedures. M-TRACE assigns Project ID: MPLADS-MH-XXXX.
- **Execution:** Contractor Bhushan handles execution, submitting progress reports, geo-tagged photos, and GST invoices via the mobile application.

- **AI Analysis:** M-TRACE detects multiple signals:
 - Signal 1: Project is significantly behind schedule.
 - Signal 2: Expenditure is unusually high relative to physical progress.
 - Signal 3: Submitted photograph exhibits a GPS location mismatch.
 - Signal 4: Material invoice prices differ significantly from benchmarks.
 - Signal 5: Similar project records appear nearby.
- **Risk Generation:** M-TRACE assigns a High Risk Score of **84/100 (● Red)**, providing specific reasons and recommending a review of invoices and field verification.
- **Administrative Resolution:** The district authority reviews the case. If legitimate (e.g., additional foundation work due to soil conditions), the reason is recorded and the case is resolved. If unjustified, administrative action is initiated through existing procedures.

15. Expected Impact & Strategic Value

1. **Early Detection:** Problems are identified during project execution rather than post-completion.
2. **Reduced Manual Burden:** AI enables authorities to prioritize high-risk cases efficiently.
3. **Enhanced Financial Oversight:** Unusual expenditure and cost deviations are flagged instantly.
4. **Improved Progress Visibility:** Comprehensive tracking of delays and physical milestones.
5. **Geospatial & Evidence Control:** GIS mapping and geo-tagged photo verification prevent location fraud.
6. **Data-Driven Accountability:** Immutable audit trails ensure complete traceability of administrative actions.
7. **Scalable National Governance:** Architecture supports seamless monitoring across all states, districts, and constituencies.

16. Conclusion & One-Line Pitch

M-TRACE transforms MPLADS monitoring from periodic, largely manual examination into continuous, risk-based, and evidence-driven intelligence. By bridging the gap between massive project data and human oversight, M-TRACE empowers government authorities to ensure transparent, accountable, and efficient asset creation across India.

The One-Line Pitch for SIH 2026:

“M-TRACE turns every MPLADS project into a continuously monitored, evidence-backed and risk-intelligent project.”